

Homework 06: Feature Matching; Hough Transform; Optical Flow

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Handout: 2025-10-06

Due: 2025-10-13, 11:59pm, on Canvas

General Instructions:

- You should solve the homework and submit your report **individually**. Identical submissions will receive a grade of zero.
- Getting help from others or checking your answers with other students (not the TAs) is okay and encouraged.
- Ask any questions on **Ed Discussion** (instead of emailing).
- **Before** the homework due date, TAs are strictly prohibited from **pre-grading** your homework. Do not expect the TAs to help you verify if your answers are correct or give you the problem solution.
- **After** the homework due date, if you do not know how to solve a problem, reach out to the TAs. They will walk you through the solution and help you understand it. Note that homework solutions will **not** be posted because some problems will be used in next year's class.
- **Exams** may contain questions related to homework, so make sure you learn how to solve the homework problems correctly.
- The deliverables are outlined for each problem, and you should carefully **follow the instructions**. Failing to follow instructions will result in **points being subtracted**.
- You will submit a **single PDF** file to Canvas as your homework report. The PDF must contain your **answers** and any requested **outputs** (e.g., printouts, snapshots of code, or GUIs). If requested, follow the instructions specified by the problem to provide your **code** (e.g., in a compressed .zip or .tar file) in addition to the PDF file.
- **Grading:** Each homework in this class will contribute **5pts** to your final grade (there will be 12 homework assignments, each 5pts, leading to 60pts for all assignments). A detailed grading **rubric** will be posted on **Canvas** after the homework due date. Any bonus points will be added to your overall course bonus points, which will be added to your final grade.
- **Late submission:** Late or missed submission will not be accepted and will receive a grade a zero. Any excused absence must be documented and disclosed to the instructor (extensions will be granted on a case-by-case basis). Three or more missed homework lead to an INC grade.

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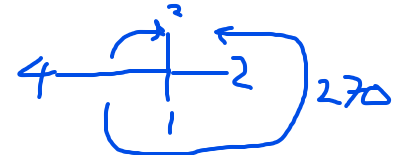
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Student: Duan Nguyen, **Date:** Oct 10th 2025

EXERCISE 1 (2pts) – The objective of this problem is to develop a high-level understanding of the SIFT algorithm. Given a keypoint at the center of a 3x3 pixel window on the image, you should compute a descriptor for the keypoint by following the instructions below. The magnitude and direction (in degrees) of the gradient for each pixel in the window is given below.

Grad_magnitude =	1	0	1
	1	0	1
	1	1	0

Grad_angle =	100	20	300
	300	100	300
	300	20	200



Step 1: Similar to SIFT algorithm, estimate the orientation of the keypoint via creating a histogram from gradient directions in the 3x3 window. Use bins $[0, 90]$, $[90, 180]$, $[180, 270]$, $[270, 360]$ for the histogram. Report the histogram values in the table below:

Histogram bin	$[0, 90]$	$[90, 180]$	$[180, 270]$	$[270, 360]$
Histogram value	2	2	1	4



Step 2: Normalize the orientation by rotating the 3x3 window using either a rotation of **0, 90, 180, or 270** degrees. That is, the orientation histogram of the rotated window must have the largest value in the bin $[0, 90]$. Report the **rotation value**, and the orientation histogram **after the rotation** in the following table: **either 90 degrees clockwise or 270 degrees counterclockwise**

Histogram bin	$[0, 90]$	$[90, 180]$	$[180, 270]$	$[270, 360]$
Histogram value	4	2	2	1

In addition, report the gradient **magnitude** and **angle** of the **rotated window** in the following 3x3 table:

Grad_magnitude =	1	1	1
	1	0	0
	0	1	1

Grad_angle =	30	30	190
	110	190	110
	290	30	30

$300 + (+90 \text{ or } -270) = 30$

$200 + (+90 \text{ or } -270) = 290$

$20 + (+90 \text{ or } -270) = 110$

$100 + (+90 \text{ or } -270) = 190$

Note that all angles must be in the $[0, 360]$ range.

Step 3: Similar to SIFT algorithm, use the rotated window gradient to construct a descriptor vector from the sum of gradient magnitudes. Use bins $[0, 90]$, $[90, 180]$, $[180, 270]$, $[270, 360]$ for the angle histogram, and report the descriptor vector in the following table:

Histogram bin	$[0, 90]$	$[90, 180]$	$[180, 270]$	$[270, 360]$
Descriptor value	4	1	1	0

Deliverables: All tables and values requested in Steps 1-3.

30	30	190
110	0	0
0	30	30

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EXERCISE 2 (3pts) – The objective of this problem is to understand how feature points are matched using their descriptors. Consider keypoints k_1 and k_2 in image I with descriptor vectors respectively as

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

On image I' , we have keypoints k'_1, k'_2 and k'_3 , with descriptor vectors respectively as

$$v'_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, v'_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, v'_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Step 1: Using **squared** Euclidean norm as distance, i.e., $d(v, v') = \|v - v'\|^2$, compute the pairwise distance between all descriptors and report the results in the table below:

Pairwise descriptor distance	v'_1	v'_2	v'_3
v_1	2	3	1
v_2	3	2	2

Step 2: Match each keypoint in image I to a keypoint in image I' based on closest descriptor. If there is a tie, either match is acceptable. Prune matches by computing the Low's ratio and removing bad matches using a threshold of **0.7**. Report the results in the following table (use “none” for pruned matches):

keypoint in image I	Matched keypoint in I' before ratio test	Low's ratio	Matched keypoint in I' after ratio test
k_1	k'3	$1/2=0.5 < 0.7$	k'3
k_2	k'2, k'3	$2/2=1 > 0.7$	none

Step 3: Similar to Step 2, match each keypoint in image I' to a keypoint in image I based on closest descriptor. Report the results in the following table:

keypoint in image I'	Matched keypoint in I before ratio test	Low's ratio	Matched keypoint in I after ratio test
k'_1	k1	$2/3=0.66... < 0.7$	k1
k'_2	k2	$2/3=0.66... < 0.7$	k2
k'_3	k1	$1/2=0.5 < 0.7$	k1

Step 4: Are the matches bidirectional? That is, if keypoint k is matched to keypoint k' , does k' also matches to k ? If yes, is that always the case? Given a set of keypoint matches from image I to I' and vice versa, how can we select matches that are bidirectional?

Deliverables: Completed tables for steps 1-3 and answer to questions for step 4.

The matches are kind of bidirectional since is it not always the case since k2 does not match to an after pruning, and only k1 and k'3 match bidirectionally since in step 3, both k'1 and k'3 match with only k'3 matches bidirectionally with k1 since k1 matches k'3 in step 2.